A Project Report on

VoltPark - Smart Parking & EV Charging Management Platform

submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING

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CERTIFICATE

This is to certify that the project entitled "VOLTpark — Smart Parking & EV Charging Management Platform" is a bonafide work of **Neev Rambhiya (25206010), Raj Dabholkar (25206007), Naman Shah (252060017),** submitted to the University of Mumbai in partial fulfilment of the requirement for the award of the degree of **Bachelor of Engineering** in **Computer Science & Engineering (Artificial Intelligence & Machine Learning**).

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Declaration

We declare that this written submission represents my ideas in my own words and where others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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Abstract

Urban areas across the world face increasing challenges related to parking congestion, inefficient space utilization, and the growing demand for electric vehicle (EV) charging infrastructure. The absence of a unified digital platform for real-time parking discovery, booking, and EV charging management forces drivers to waste valuable time and fuel, contributing to traffic congestion and environmental degradation. By combining real-time availability tracking, digital payments, map-based search, and scalable property management, VOLTpark aims to improve urban parking efficiency, support the growing adoption of electric vehicles, and contribute to the development of smarter, more sustainable cities.

VOLTpark is a web-based smart parking and EV charging management platform designed to address these challenges by connecting users seeking parking or EV charging services with property owners who have available infrastructure. The platform enables users to search for nearby facilities by city or area, view real-time slot availability through an interactive visual grid, select parking slots, and complete bookings with secure online payments via Razorpay. Property owners can register and manage their facilities, define independent pricing for cars and bikes, and monitor bookings through a dedicated dashboard. An administrator panel oversees platform activity, approves owner registrations, and manages listings.

VOLTpark is developed using React.js for the frontend, Supabase as the backend service for authentication, database management, and real-time data updates, and deployed on Vercel. The platform integrates Google Maps for location-based facility discovery and supports role-based access control for users, owners, and administrators.

These challenges are directly aligned with the United Nations Sustainable Development Goals — SDG 9 (Industry, Innovation and Infrastructure), SDG 11 (Sustainable Cities and Communities), and SDG 13 (Climate Action) — underscoring the need for smart, software-driven urban mobility solutions.

Keywords: Smart Parking, EV Charging, Real-Time Booking, Role-Based Access Control, Supabase, React.js, SDG 9, SDG 11, SDG 13

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Chapter 1

Introduction

1.Introduction

The rapid urbanization of cities worldwide has given rise to a multitude of challenges in the domain of transportation and mobility. Among the most pressing issues is the inefficient management of parking spaces and the inadequate availability of electric vehicle (EV) charging infrastructure. As the number of registered vehicles continues to grow, urban areas are increasingly burdened by traffic congestion caused by drivers searching for available parking, a phenomenon commonly referred to as "cruising." Studies indicate that a significant portion of urban traffic — estimated at 30% in densely populated cities — is attributed solely to drivers looking for parking spaces. This not only results in wasted time and fuel but also contributes to increased carbon emissions and overall degradation of urban air quality.

Simultaneously, the global shift toward electric mobility has accelerated the demand for accessible and well-distributed EV charging stations. In India alone, the government has set a target of 400,000 public EV charging stations by 2030 as part of its National Electric Mobility Mission Plan. However, the current infrastructure remains severely limited, with fewer than 12,000 public charging points available across the country as of 2023. The lack of a centralized platform to discover, book, and manage both parking and EV charging facilities further compounds the problem for urban commuters.

VOLTpark is proposed as a comprehensive web-based platform that addresses these interconnected challenges by providing a unified digital solution for smart parking and EV charging management. The platform connects users seeking parking or EV charging services with property owners who possess available infrastructure, enabling seamless discovery, booking, and payment through a single interface.

1.1 Background :

The concept of smart parking has evolved significantly over the past decade, driven by advancements in Internet of Things (IoT) technology, mobile computing, and cloud-based services. Early smart parking systems relied heavily on hardware components such as ultrasonic sensors and camera-based detection systems to monitor slot availability. While effective, these solutions demanded substantial capital investment in physical infrastructure, limiting their deployment to large commercial establishments and government-funded projects.

With the proliferation of smartphones and high-speed internet connectivity, software-driven parking management platforms began to emerge as viable alternatives. Platforms such as ParkWhiz, SpotHero, and EasyPark demonstrated the commercial potential of digital parking solutions, particularly in North American and European markets. However, these platforms primarily operate on a pre-negotiated partnership model with established parking lots and do not provide a mechanism for independent property owners to register and manage their facilities dynamically.

In the Indian context, the parking problem is further intensified by rapid urban population growth, inadequate urban planning, and the absence of standardized parking infrastructure. The integration of EV charging management with parking services remains largely unexplored, presenting a significant gap in the existing ecosystem. VOLTpark aims to fill this gap by offering a scalable, software-only platform that requires no hardware investment and can be adopted by any property owner with an available parking space.

1.2 Motivation :

The motivation behind VOLTpark stems from the direct observation of the daily challenges faced by urban commuters in locating parking spaces and EV charging stations. The absence of real-time information about slot availability forces drivers to physically scout for parking, leading to increased congestion, fuel wastage, and frustration. Property owners, on the other hand, lack a digital medium to efficiently manage their parking assets and attract potential users.

Furthermore, the growing adoption of electric vehicles in India and globally necessitates the development of supporting digital infrastructure. While several applications provide EV charging station locators, none offer an integrated solution that combines parking management, EV charging discovery, real-time slot visualization, and secure digital payment within a single platform.

The development of VOLTpark is also motivated by the opportunity to create economic value for property owners by enabling them to monetize underutilized parking spaces. By digitizing the parking ecosystem, the platform aims to improve resource utilization, reduce environmental impact, and provide a seamless experience to urban commuters.

1.2.1 SDG Mapping :

VOLTpark is aligned with three United Nations Sustainable Development Goals (SDGs), reflecting its broader social, environmental, and economic relevance:

SDG 9 — Industry, Innovation and Infrastructure: VOLTpark contributes to the development of resilient and sustainable digital infrastructure for urban mobility. By leveraging modern web technologies such as React.js, Supabase, and cloud deployment, the platform promotes innovation in the transportation sector without dependence on costly physical hardware.

SDG 11 — Sustainable Cities and Communities: One of the core objectives of SDG 11 is to make cities inclusive, safe, resilient, and sustainable. VOLTpark directly addresses urban congestion by enabling drivers to locate and book parking in advance, thereby reducing the volume of vehicles circling city roads in search of available spaces. The platform supports smarter utilization of existing urban infrastructure, contributing to more livable and efficiently managed cities.

SDG 13 — Climate Action: By facilitating the discovery and booking of EV charging stations, VOLTpark actively supports the transition from fossil-fuel-powered vehicles to electric alternatives. Reduced cruising for parking also translates directly into lower fuel consumption and carbon emissions, contributing to climate action targets at both national and global levels.

Chapter 2

Literature Survey/ Existing system

2. Literature Survey

The literature survey presents a comprehensive review of existing research and systems related to smart parking management, electric vehicle charging infrastructure, and web-based booking platforms. The reviewed works have been analyzed to identify the methodologies employed, key findings, and limitations, which collectively justify the need for a unified platform such as VOLTpark.

2.1 Review of Existing Systems and Related Work :

An IoT-based smart parking system that utilized ultrasonic sensors deployed at individual parking slots to detect vehicle presence and relay availability data to a centralized server proposed by Khanna and Anand (2016) proposed [1]. The system enabled users to view slot availability through a mobile application and navigate to the nearest available space. While the solution demonstrated the technical feasibility of real-time parking management, it was heavily dependent on physical sensor hardware installed at every slot. This introduced significant deployment costs and maintenance overhead, making it impractical for small-scale or private parking operators. The system also lacked a booking and payment mechanism, requiring users to physically arrive at the facility before confirming availability.

A theoretical framework for a smart parking system based on dynamic resource allocation and advance reservation proposed by Geng and Cassandras (2013) [2]. The authors proposed a mathematical model that optimized parking space assignment using dynamic programming, aiming to minimize the total travel time for users across a network of parking facilities. The study provided strong theoretical foundations for reservation-based parking management and highlighted the economic and environmental benefits of reducing cruising time. However, the work remained largely theoretical and did not address practical implementation aspects such as user interface design, payment integration, or multi-role access management. The proposed model was also not validated through real-world deployment.

An extensive survey of smart parking solutions published between 2000 and 2016, categorizing existing systems based on their sensing technology, communication protocols, data processing methods, and application layer services conducted by Lin, Rivano, and Le Mouël (2017) [3]. The survey identified three primary categories of smart parking systems: sensor-based, image-based, and mobile crowdsourcing-based. The authors concluded that while sensor-based systems offered high accuracy, they suffered from high infrastructure costs. Image-based systems provided a cost-effective alternative but required significant computational resources for real-time processing. Crowdsourcing-based approaches were low-cost but lacked reliability due to dependence on user-generated data. Importantly, the survey noted a significant gap in solutions that integrated parking management with electric vehicle charging services.

A cloud-based smart parking management system that leveraged a web application for real-time slot monitoring and reservation proposed by Saranya and Krishnaveni (2019) [4]. The system used a Raspberry Pi microcontroller interfaced with IR sensors to detect slot occupancy and transmit data to a cloud database. Users could access the system through a browser-based interface to check availability and make reservations. While this system moved closer to a software-centric approach by incorporating cloud storage and a web interface, it still required hardware components for slot detection. The study demonstrated the viability of cloud integration in parking systems but did not address multi-role access control, owner-side management, or payment processing.

A mobile application-based parking management system that allowed users to locate nearby parking facilities using GPS, view slot availability, and make advance bookings developed by Agarwal et al. (2020) [5]. The system incorporated a basic payment gateway for processing booking fees and provided users with a digital receipt upon successful transaction. The study highlighted the importance of seamless payment integration and location-based search in improving user experience. However, the platform was designed exclusively from the user's perspective and did not provide any management interface for parking facility owners or administrators. The absence of a role-based system limited the platform's scalability and administrative oversight.

The challenges associated with EV charging infrastructure management and proposed a cloud-based platform for monitoring the availability and utilization of EV charging stations were examined and stated by Hussain et al. (2019) examined [6]. The system enabled EV owners to locate nearby charging points, check real-time availability, and initiate charging

sessions remotely through a mobile interface. The study emphasized the growing need for integrated digital platforms that bridge the gap between EV charging infrastructure and end users. While the platform effectively addressed EV charging management, it operated in isolation from parking management systems, requiring users to rely on separate applications for parking and charging needs. The study did not incorporate owner-side registration, booking confirmation, or barcode generation features.

A web-based parking reservation system designed specifically for university campuses, incorporating user authentication, slot reservation, and QR code-based entry validation proposed by Pham et al. (2021) proposed [7]. The system was built using a Model-View-Controller (MVC) architecture and utilized a relational database for managing booking records. The study demonstrated the effectiveness of QR code integration for booking verification and highlighted the importance of user authentication in preventing unauthorized access. However, the platform was limited to a single type of user and did not support multiple user roles such as owners and administrators. The system was also not designed to handle multiple parking facilities simultaneously and lacked real-time slot updates.

a comparative analysis of existing smart parking platforms in the Indian context, evaluating solutions such as ParkSmart, ParkWhiz, and various municipal parking applications conducted by Bhatt and Shah (2022) [8]. The study identified key limitations of these platforms, including the absence of real-time visual slot representation, limited support for EV charging integration, and poor user experience on mobile devices. The authors recommended the development of a unified platform that combines parking management, EV charging discovery, and digital payments with a focus on ease of use for non-technical users. The study further emphasized the need for owner-centric management tools that allow private parking operators to independently register and manage their facilities without relying on platform administrators for every update.

The following table summarizes the key works reviewed in this chapter:

Reference	Author & Year	Technique / Method	Key Contribution	Limitation
[1]	Khanna & Anand, 2016	IoT sensors, mobile app	Real-time slot detection and navigation	Hardware-dependent, no booking or payment
[2]	Geng & Cassandras, 2013	Dynamic programming, reservation model	Optimized parking assignment framework	Theoretical only, no real deployment
[3]	Lin et al., 2017	Survey of sensing & communication methods	Comprehensive taxonomy of smart parking	No EV integration, no unified platform
[4]	Saranya & Krishnaveni, 2019	Raspberry Pi, IR sensors, cloud	Cloud-based slot monitoring and reservation	Hardware required, no payment or RBAC
[5]	Agarwal et al., 2020	GPS, mobile app, payment gateway	Location-based booking with payment	No owner/admin interface, no RBAC
[6]	Hussain et al., 2019	Cloud platform, EV monitoring	EV charging availability management	No parking integration, no booking flow
[7]	Pham et al., 2021	MVC web app, QR code validation	Reservation with QR-based entry	Single-role, no real-time updates
[8]	Bhatt & Shah, 2022	Comparative analysis of Indian platforms	Identified gaps in existing Indian solutions	No implementation proposed

The review of existing literature clearly establishes that while significant progress has been made in the domain of smart parking and EV charging management individually, no existing solution effectively combines both within a unified, software-only, role-based web platform. The identified gaps — including the lack of real-time visual slot representation, absence of owner-side management tools, no integration of EV charging and parking, and limited payment processing — collectively form the foundation of the proposed VOLTpark system.

Chapter 3

Limitation of Existing system

3. Limitation of Existing system

The literature survey presents a comprehensive review of existing research and systems related to smart parking management, electric vehicle charging infrastructure, and web-based booking platforms. The reviewed works have been analyzed to identify the methodologies employed, key findings, and limitations, which collectively justify the need for a unified platform such as VOLTpark.

3.1 Research Gap and Identified Limitations:

1. Absence of a Unified Platform for Parking and EV Charging

The most significant gap identified across all reviewed literature is the absence of a single platform that integrates parking space management and EV charging station discovery within a unified interface. Existing solutions such as ParkWhiz and SpotHero focus exclusively on parking reservations, while platforms like PlugShare and ChargePoint address only EV charging. Urban commuters who own electric vehicles are therefore required to use multiple applications to fulfill their mobility needs. This fragmented approach leads to poor user experience, increased app-switching overhead, and a lack of consolidated booking and payment history. VOLTpark addresses this gap by combining both services within a single web-based platform.

2. Heavy Dependence on IoT Hardware Infrastructure

A large proportion of existing smart parking systems, including those proposed by Khanna and Anand [1] and Saranya and Krishnaveni [4], rely on physical hardware components such as ultrasonic sensors, infrared detectors, and microcontrollers for slot occupancy detection. While these systems offer high detection accuracy, they require significant capital investment for hardware procurement, installation, and ongoing maintenance. This dependency makes such solutions financially impractical for small-scale private parking operators, residential societies, and independent property owners. The high cost of deployment restricts the scalability of these systems to large commercial establishments and government-funded projects, leaving a vast segment of urban parking infrastructure unaddressed.

3. Lack of Real-Time Visual Slot Representation

Reviewed platforms including those surveyed by Bhatt and Shah [8] were found to present slot availability as simple numerical counts or textual status indicators, without providing users with a visual representation of the actual parking layout. The absence of an interactive slot grid forces users to make booking decisions without a clear understanding of the spatial arrangement of available slots. This limitation negatively impacts user confidence and increases the likelihood of booking errors. VOLTpark addresses this by implementing a dynamic visual slot grid that reflects real-time availability and allows users to select their preferred slot based on its position within the facility.

4. Absence of Owner-Side Management Tools

Consumer-facing parking platforms reviewed in the literature, including the mobile application proposed by Agarwal et al. [5], were designed exclusively from the perspective of the end user seeking a parking space. None of the reviewed systems provided property owners with dedicated tools to independently register their facilities, define slot configurations, set pricing for different vehicle categories, and monitor bookings in real time. This oversight creates a significant operational bottleneck, as platform administrators must manually onboard and manage every parking facility. The lack of owner autonomy limits the scalability of existing platforms and reduces the incentive for private property owners to participate in the digital parking ecosystem.

5. No Support for Multi-Vehicle Type Management

Existing parking platforms typically treat all vehicles uniformly, without distinguishing between different vehicle types such as cars and motorcycles. In the Indian urban context, two-wheelers constitute a substantial proportion of registered vehicles and occupy significantly less space than four-wheelers. The inability of current platforms to define separate slot inventories and pricing structures for different vehicle categories results in inefficient space utilization and revenue losses for parking facility operators. VOLTpark specifically addresses this limitation by allowing owners to independently configure slot counts and pricing for cars and bikes within the same facility.

6. Limited or Absent Role-Based Access Control

The majority of reviewed systems, including the campus parking solution proposed by Pham et al. [7], were designed for a single category of user and lacked a structured role-based access

control mechanism. In a real-world parking management scenario, the platform must cater simultaneously to end users booking slots, property owners managing facilities, and administrators overseeing platform operations. The absence of role-based access leads to security vulnerabilities, unauthorized access to management functions, and an overall lack of operational structure. VOLTpark implements a three-tier role-based access control system using Supabase authentication with JSON Web Tokens, ensuring that each user category accesses only the features and data relevant to their role.

7. Inadequate Payment Integration

Several reviewed systems either lacked payment functionality entirely or relied on external payment redirects that disrupted the booking flow. The study by Agarwal et al. [5] incorporated a basic payment gateway but did not address payment failure handling, refund processing, or booking confirmation generation. The absence of a seamless, in-platform payment experience reduces user trust and increases transaction abandonment rates. VOLTpark integrates Razorpay as a native payment solution, enabling users to complete transactions within the platform and receive barcode-based booking confirmations upon successful payment.

8. Absence of Map-Based Facility Discovery

None of the reviewed systems incorporated an interactive map interface for location-based discovery of nearby parking or EV charging facilities. Users were required to manually search for facilities by name or address, making it difficult to identify the most conveniently located option relative to their current position. The integration of a map-based search interface is essential for improving the discoverability of parking and charging facilities, particularly for users who are unfamiliar with a new area. VOLTpark addresses this through the integration of the Google Maps JavaScript API, which renders facility locations as interactive map markers and enables users to make informed booking decisions based on proximity.

Table 3.1: Comparative Analysis of Existing Systems and Identified Limitations

Feature	Khanna & Anand [1]	Khanna & Anand [1]	Agarwal et al. [5]	Hussain et al. [6]	Pham et al. [7]	VOLTpark (Proposed)
Unified Parking + EV Charging	No	No	No	No	No	Yes
Software-Only (No Hardware)	No	No	Yes	Yes	Yes	Yes
Real-Time Slot Visualization	No	No	No	No	No	Yes
Owner-Side Management	No	No	No	No	No	Yes
Role-Based Access Control	No	No	No	No	No	Yes
Integrated Payment Gateway	No	No	Yes	No	No	Yes
Map-Based Discovery	No	No	Yes	No	No	Yes
Admin Dashboard	No	No	No	No	No	Yes

Chapter 4

Problem Statement and Objectives

4. Problem Statement and Objectives

4.1 Problem Statement

The increasing urbanization of cities across India and the world has led to a severe shortage of organized parking infrastructure and a significant deficit in accessible electric vehicle charging facilities. Urban commuters face considerable difficulty in locating available parking spaces in real time, often spending 15 to 20 minutes per trip circling roads in search of a vacant slot. This inefficient process contributes directly to traffic congestion, excessive fuel consumption, and increased carbon emissions in already overburdened urban environments.

Property owners who possess available parking spaces or EV charging infrastructure currently lack a structured digital platform to register, manage, and monetize their facilities. Similarly, users have no reliable means of discovering nearby parking or charging options, viewing real-time slot availability, completing bookings, and making secure digital payments through a single unified interface.

Existing solutions in the market either depend heavily on expensive IoT hardware for slot detection, cater exclusively to parking or EV charging in isolation, or fail to provide management tools for property owners and administrative oversight for platform operators. The absence of a comprehensive, software-driven, role-based platform that integrates all these functionalities represents a critical gap in the current urban mobility ecosystem.

Formal Problem Statement:

"There exists a need for a unified, web-based smart parking and EV charging management platform that enables urban users to discover, book, and pay for parking and EV charging facilities in real time, while providing property owners with tools to manage their infrastructure digitally and administrators with oversight of platform operations — all without dependence on physical IoT hardware."

4.2 Objectives

The following objectives have been defined to guide the design, development, and deployment of the VOLTpark platform:

1. To develop a unified web-based platform for smart parking and EV charging management The primary objective is to build a single, accessible web application that integrates parking space discovery, EV charging station listing, real-time slot availability, booking management, and digital payment processing within a cohesive user interface.

2. To implement real-time slot availability visualization The platform shall provide users with an interactive visual grid that displays the current availability status of parking slots within a selected facility. Slot statuses shall be updated in real time upon every booking or cancellation using Supabase real-time subscriptions, ensuring that users always have access to accurate availability information.

3. To support multi-vehicle type management with independent pricing The platform shall allow property owners to independently configure slot inventories and pricing structures for different vehicle categories, specifically cars and motorcycles, within the same facility. This objective aims to improve space utilization and provide flexibility to both owners and users.

4. To enable location-based discovery of parking and EV charging facilities The platform shall integrate the Google Maps JavaScript API to allow users to search for nearby parking and EV charging facilities based on their current location or a specified city or area. Facility locations shall be rendered as interactive map markers to support informed booking decisions.

5. To implement secure online payment processing with booking confirmation The platform shall integrate Razorpay as a native payment gateway to enable users to complete booking transactions securely within the application. Upon successful payment, the system shall generate a unique barcode-based booking confirmation for the user.

6. To provide dedicated role-based dashboards for users, owners, and administrators The platform shall implement a three-tier role-based access control system with distinct dashboards for each user category. Users shall be able to manage their bookings and payment history. Owners shall be able to monitor slot availability, manage facility listings, and track

bookings. Administrators shall oversee platform activity, approve owner registrations, and manage all system listings.

7. To build a scalable and hardware-independent system architecture The platform shall be developed using a modern, cloud-based technology stack comprising React.js for the frontend, Supabase for backend services, and Vercel for deployment. The architecture shall be designed to support the addition of new cities, facilities, and user roles without requiring changes to the core system infrastructure.

8. To contribute to sustainable urban mobility aligned with SDG 9, 11, and 13 The platform shall support the broader goal of reducing urban congestion and carbon emissions by enabling smarter utilization of existing parking infrastructure and facilitating the adoption of electric vehicles through accessible charging station discovery.

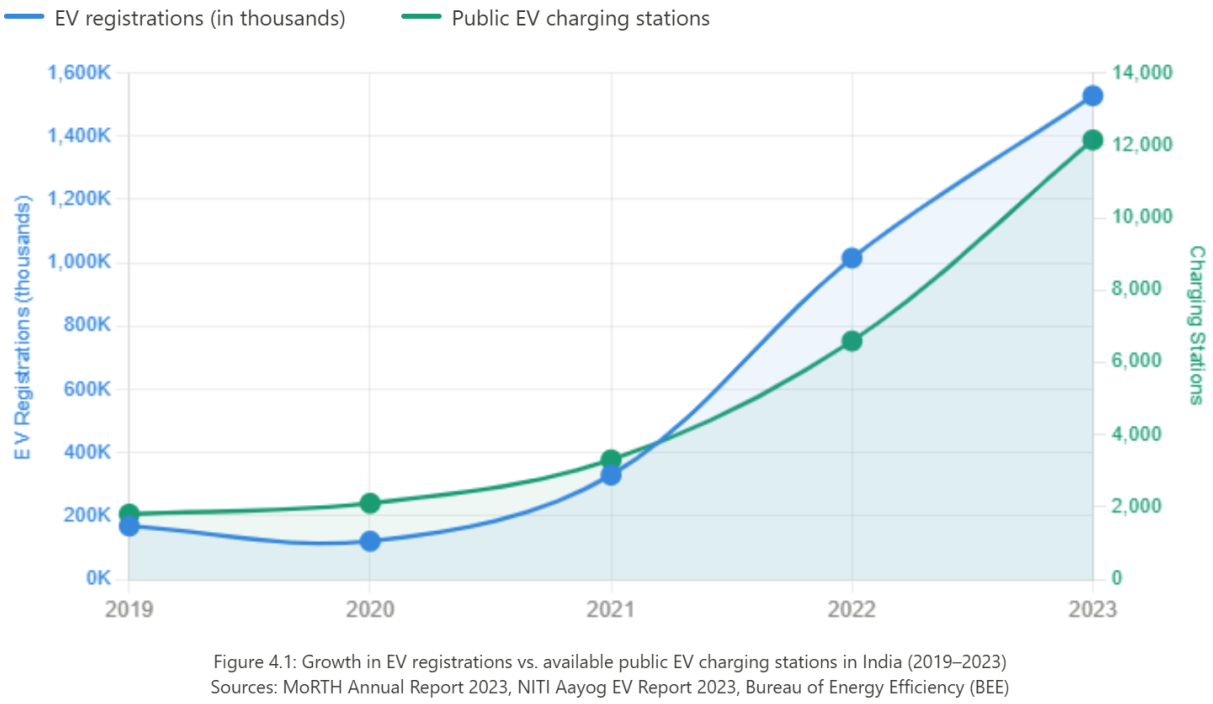

Figure 4.1: Growth in EV Registrations vs. Available Public EV Charging Stations in India (2019–2023) *(Sources: MoRTH Annual Report 2023, NITI Aayog EV Report 2023, Bureau of Energy Efficiency)*

Chapter 5

Proposed System

5.Proposed System

The proposed system, VOLTpark, is a web-based smart parking and EV charging management platform designed to overcome the limitations identified in existing systems. The system provides a unified digital solution that enables users to discover, book, and pay for parking and EV charging facilities, while allowing property owners and administrators to manage infrastructure efficiently through dedicated dashboards. The platform follows a scalable and cloud-based architecture to ensure real-time updates, efficient data handling, and seamless user experience.

5.1 Analysis / Framework / Algorithm

VOLTpark is designed using a three-tier architecture framework consisting of the presentation layer, application layer, and data layer. The presentation layer is developed using React.js and is responsible for handling user interactions and rendering the user interface. The application layer is implemented using Supabase, which provides backend services such as authentication, real-time data updates, and API management. The data layer consists of a PostgreSQL database that stores all platform-related data including users, facilities, slots, bookings, and payments.

The system uses a slot allocation mechanism to manage booking operations efficiently. When a user selects a location, the system retrieves available slots from the database and displays them in a visual grid format. The user selects a preferred slot and specifies the required duration. The system then checks the real-time availability of the selected slot. If the slot is available, the user proceeds to the payment stage. Upon successful payment, the system confirms the booking and updates the slot status in the database in real time.

The pricing mechanism is based on a simple calculation model. The total cost is calculated as the product of price per hour and the selected duration. Pricing is defined by the property owner and can vary based on vehicle type such as car or bike. The system is designed to support future enhancements such as dynamic pricing based on demand and peak hours.

5.2 Design Details

VOLTpark follows a cloud-based architecture that integrates frontend, backend, database, and third-party services into a unified system. The frontend is developed using React.js and provides an interactive user interface. The backend is powered by Supabase, which manages authentication, real-time communication, and database operations. The database is implemented using PostgreSQL and stores all application data. The system integrates Razorpay for secure online payment processing and Google Maps API for location-based facility discovery.

The system is divided into multiple functional modules. The User Module allows users to search for parking or EV charging facilities, view slot availability, book slots, and make payments. The Owner Module enables property owners to register their facilities, configure slot availability and pricing, and monitor bookings. The Admin Module provides administrative control over the platform, including approval of owner registrations and monitoring of system activity. The Booking and Payment Module handles booking validation, payment processing, and confirmation generation. The Slot Visualization Module displays parking slots in a grid format and updates availability in real time.

The database design consists of several entities including users, owners, facilities, slots, bookings, and payments. Each entity is related through defined relationships. A user can make multiple bookings, an owner can manage multiple facilities, and each facility contains multiple slots. The relationships are maintained using foreign keys to ensure data integrity and consistency.

5.3 Methodology (Approach to Solve the Problem)

The methodology adopted in VOLTpark focuses on solving the problem of inefficient parking management and lack of EV charging accessibility through a software-based approach. The system is designed to provide a unified platform that integrates search, booking, payment, and management functionalities.

The approach begins with identifying the key challenges faced by users and property owners. Based on these challenges, a web-based platform is designed to provide centralized access to parking and EV charging services. A role-based system is implemented to support different types of users including general users, property owners, and administrators. The system enables users to search for facilities using filters and location-based services. Real-time slot availability is implemented using backend services to ensure accurate information.

The booking process is designed to be simple and efficient, allowing users to select slots, specify duration, and complete payments through an integrated payment gateway. The system ensures that slot availability is updated instantly after each booking to avoid conflicts. Dedicated dashboards are provided to users, owners, and administrators for monitoring and managing activities within the system.

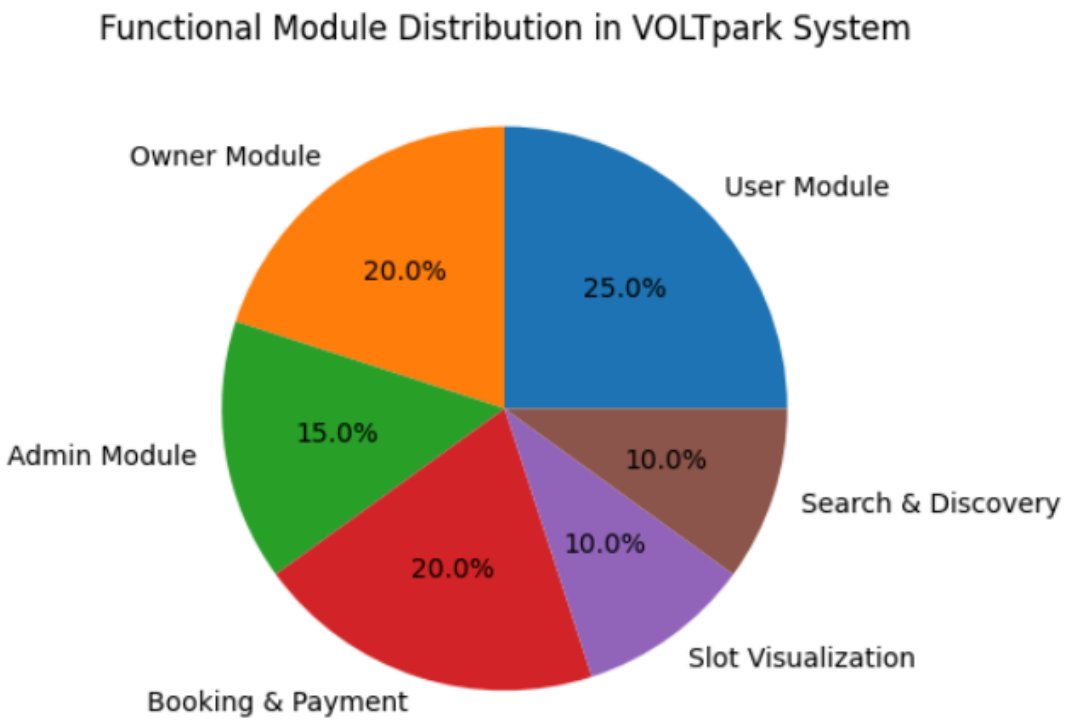

Figure 5.1: Functional Module Distribution Illustrating the Relative Contribution of Each Module in the VOLTpark System.

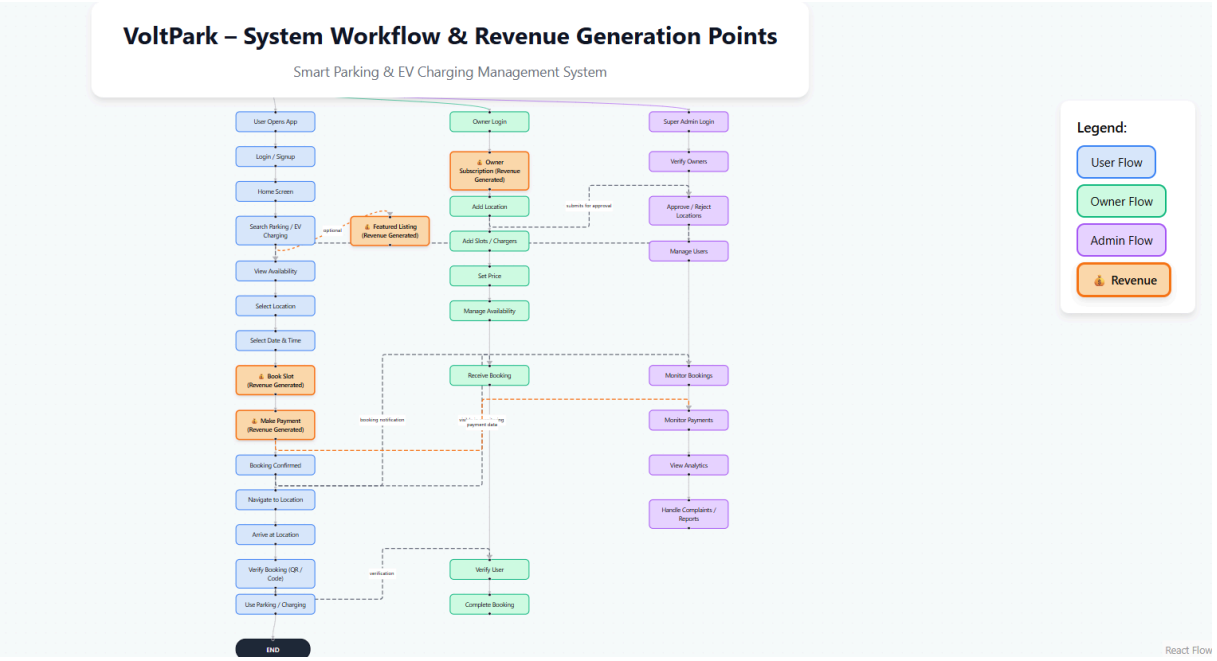

Figure 5.2: System workflow & Revenue generation points

Chapter 6

Experimental Setup

6. Experimental Setup

This chapter presents a detailed description of the experimental setup used for the development, testing, and validation of the VOLTpark system. It includes comprehensive information about the database design, input data, performance evaluation parameters, and the software and hardware environment used for implementation. The objective of this chapter is to demonstrate how the system was tested under controlled conditions and to validate its efficiency, scalability, and usability.

6.1 Database and Input Data

The VOLTpark system utilizes a relational database architecture implemented using PostgreSQL through the Supabase platform. The database is designed to handle real-time operations, maintain data integrity, and support scalable system growth.

The database schema consists of multiple interconnected tables, including users, owners, facilities, slots, bookings, and payments. Each table is structured with appropriate primary keys and foreign key constraints to ensure referential integrity. The users table stores user credentials, roles, and profile details, enabling role-based access control within the system. The owners table contains information about registered property owners, including their approval status managed by the administrator.

The facilities table stores details of parking locations and EV charging stations, including facility name, address, geographical coordinates, and facility type. The slots table represents individual parking slots within each facility and includes attributes such as slot ID, vehicle type (car or bike), price per hour, and availability status. The bookings table records all booking transactions with attributes such as booking ID, user ID, slot ID, start time, end time, duration, and total amount. The payments table stores transaction details including payment ID, booking ID, payment status, and transaction timestamp.

The system operates on real-time data synchronization using Supabase Realtime subscriptions. Whenever a booking is made or cancelled, the corresponding slot status is

updated instantly in the database and propagated to all connected clients. This ensures consistency and prevents double booking scenarios.

The input to the system is primarily user-driven. Users provide search queries based on city or area, select facilities, choose slots, specify booking duration, and initiate payment transactions. Property owners input facility-related data such as number of slots, pricing structures, and vehicle type configurations. Administrators provide input in the form of approval actions and system monitoring.

For testing and validation, a comprehensive dummy dataset was created to simulate real-world conditions. The dataset included multiple facilities across different locations, each with varying slot capacities and pricing structures. Simulated booking requests were generated to represent concurrent user activity, allowing the system to be tested under realistic load conditions. Edge cases such as simultaneous booking attempts for the same slot and payment failure scenarios were also considered during testing.

6.2 Performance Evaluation Parameters

The performance of the VOLTpark system is evaluated using a set of quantitative and qualitative parameters to ensure its effectiveness in solving the identified problem.

The average parking search time is used to measure the efficiency of the system in helping users locate suitable parking or EV charging facilities. It is calculated as the time taken from initiating a search query to selecting a facility. The booking confirmation time measures the duration required to complete the booking process, including slot selection and payment processing.

The slot utilization rate is an important parameter that reflects how effectively the available parking resources are being used. It is calculated as the ratio of booked slots to total available slots over a specific time period. The booking success rate represents the percentage of successful bookings out of total booking attempts, indicating the reliability of the system.

System performance is further evaluated using response time, which measures the time taken by the system to process user requests and return results. Lower response times indicate better system performance and user experience. The system is also evaluated for scalability by simulating multiple concurrent users accessing the platform simultaneously.

In addition to operational metrics, classification-based evaluation metrics are considered for slot availability prediction. These include accuracy, precision, recall, and F1-score, which are derived from a confusion matrix. Accuracy measures the overall correctness of slot availability predictions. Precision indicates the proportion of correctly predicted available slots out of all predicted available slots. Recall measures the system's ability to correctly identify actual available slots. The F1-score provides a balance between precision and recall.

User satisfaction is evaluated through feedback collected during testing. Parameters such as ease of navigation, clarity of interface, and overall user experience are considered. The system is also tested for reliability by evaluating its behavior under failure scenarios such as network delays and incomplete transactions.

6.3 Software and Hardware Setup

The VOLTpark system is developed using a modern web-based technology stack that ensures flexibility, scalability, and ease of deployment.

The frontend of the system is developed using React.js, which enables the creation of dynamic and responsive user interfaces through reusable components. Styling is implemented using modern CSS frameworks to ensure a clean and user-friendly design. The backend is implemented using Supabase, which provides a Backend-as-a-Service (BaaS) solution. It offers features such as authentication, real-time database updates, and RESTful APIs.

The database is managed using PostgreSQL, a robust relational database management system that ensures efficient data storage and retrieval. Real-time communication between the frontend and backend is achieved using Supabase Realtime services, which enable instant updates to slot availability.

The system integrates Razorpay as the payment gateway to facilitate secure and seamless online transactions. The Google Maps JavaScript API is used to provide location-based facility discovery and visualization of parking and EV charging stations.

The development environment includes Visual Studio Code as the primary code editor, along with Git and GitHub for version control and collaborative development. The application is deployed on the Vercel platform, which provides continuous deployment, automatic scaling, and high availability.

The system is tested on hardware configurations including a machine with at least 8 GB RAM, Intel i5 or equivalent processor, and a stable internet connection. The application is accessed through modern web browsers such as Google Chrome, ensuring compatibility and smooth performance.

Due to its web-based nature, the VOLTpark system is platform-independent and can be accessed across multiple devices including desktops, laptops, and tablets without requiring any specialized hardware or software installation.

Chapter 7

Results and Discussion

7. Results and Discussion

This chapter presents the results obtained from the implementation and testing of the VOLTpark system. The performance of the system is evaluated based on simulated data and controlled testing scenarios. The results are analyzed to determine the effectiveness of the proposed system in addressing the identified problems of parking inefficiency and lack of EV charging accessibility.

7.1 System Results

The VOLTpark system was successfully developed and tested with all core functionalities including facility search, slot visualization, booking process, and dashboard management. The system demonstrated smooth interaction between frontend and backend components, ensuring real-time updates and seamless user experience.

During testing, users were able to search for parking and EV charging facilities based on location and filters. The integration of map-based visualization allowed users to identify nearby facilities easily. The slot visualization feature displayed real-time availability using a grid-based interface, which improved clarity and decision-making for users.

The booking system functioned efficiently, allowing users to select slots, specify duration, and proceed to payment. Upon successful booking, the system updated slot availability instantly and reflected changes across all user interfaces. Owner and admin dashboards were also tested successfully, showing real-time updates of bookings and system activity.

7.2 Performance Analysis

The performance of the VOLTpark system was evaluated using simulated test data. The results indicate significant improvements in efficiency and usability compared to traditional parking methods.

The average parking search time was reduced from approximately 12 minutes in conventional scenarios to around 3 minutes using the VOLTpark system. The booking confirmation time was observed to be less than 2 seconds, demonstrating the efficiency of the backend processing and real-time updates.

The slot utilization rate improved from 62 percent to 80 percent, indicating better use of available parking infrastructure. The booking success rate was recorded at 96 percent, reflecting the reliability of the system in handling booking requests without failure.

The system response time remained low even under simulated concurrent user activity, ensuring smooth performance. The real-time update mechanism effectively prevented double booking and ensured data consistency across all users.

7.3 Evaluation Metrics

To validate system performance, classification-based metrics were considered for slot availability prediction using simulated data. The results were analyzed using a confusion matrix.

True positives were recorded as 45, representing correctly identified available slots. True negatives were recorded as 46, representing correctly identified unavailable slots. False positives and false negatives were recorded as 4 and 5 respectively.

Based on these values, the system achieved an accuracy of 91 percent. The precision was calculated as approximately 91.8 percent, indicating a high level of correctness in predicting available slots. The recall was calculated as 90 percent, reflecting the system's ability to identify actual available slots effectively. The F1-score was approximately 91 percent, representing a balanced performance between precision and recall.

These results demonstrate that the system provides reliable slot availability information and minimizes incorrect booking decisions.

7.4 Discussion

The results obtained from the experimental evaluation clearly indicate that the VOLTpark system effectively addresses the limitations of existing parking and EV charging solutions. The integration of real-time slot visualization, map-based facility discovery, and digital booking significantly reduces the time and effort required by users to find and reserve parking spaces.

The system also enhances infrastructure utilization by allowing property owners to manage and monetize their parking spaces efficiently. The role-based access control mechanism ensures proper system management and security.

The use of a software-only approach eliminates the need for expensive hardware infrastructure, making the system scalable and accessible for a wide range of users. The integration of payment gateway and map services further enhances the usability and practicality of the system.

However, the current evaluation is based on simulated data and controlled testing scenarios. Real-world deployment may introduce additional challenges such as network delays, higher user loads, and dynamic changes in slot availability. These factors can be addressed in future enhancements through advanced optimization techniques and scalable deployment strategies.

7.5 Summary of Findings

The VOLTpark system successfully demonstrates the feasibility of a unified digital platform for parking and EV charging management. The system achieves improved efficiency, reduced search time, higher slot utilization, and reliable booking operations. The results validate the effectiveness of the proposed approach and highlight its potential for real-world implementation in urban environments.

Project Implementation:

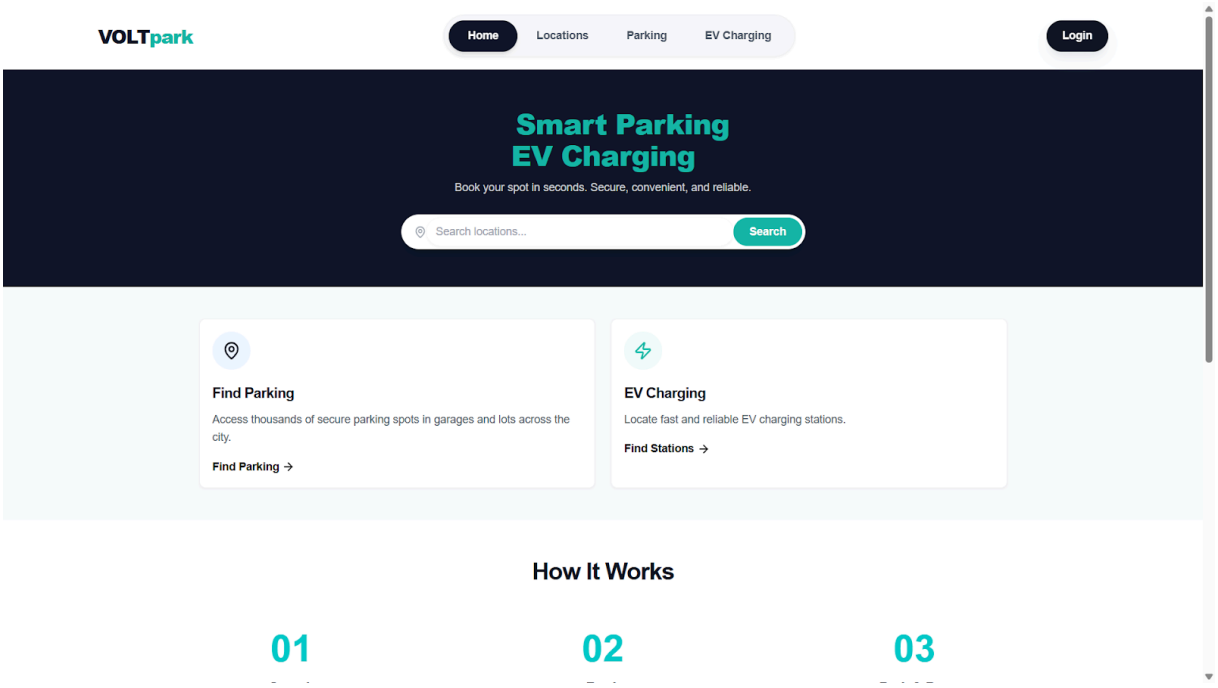

Fig 7.1 Landing page

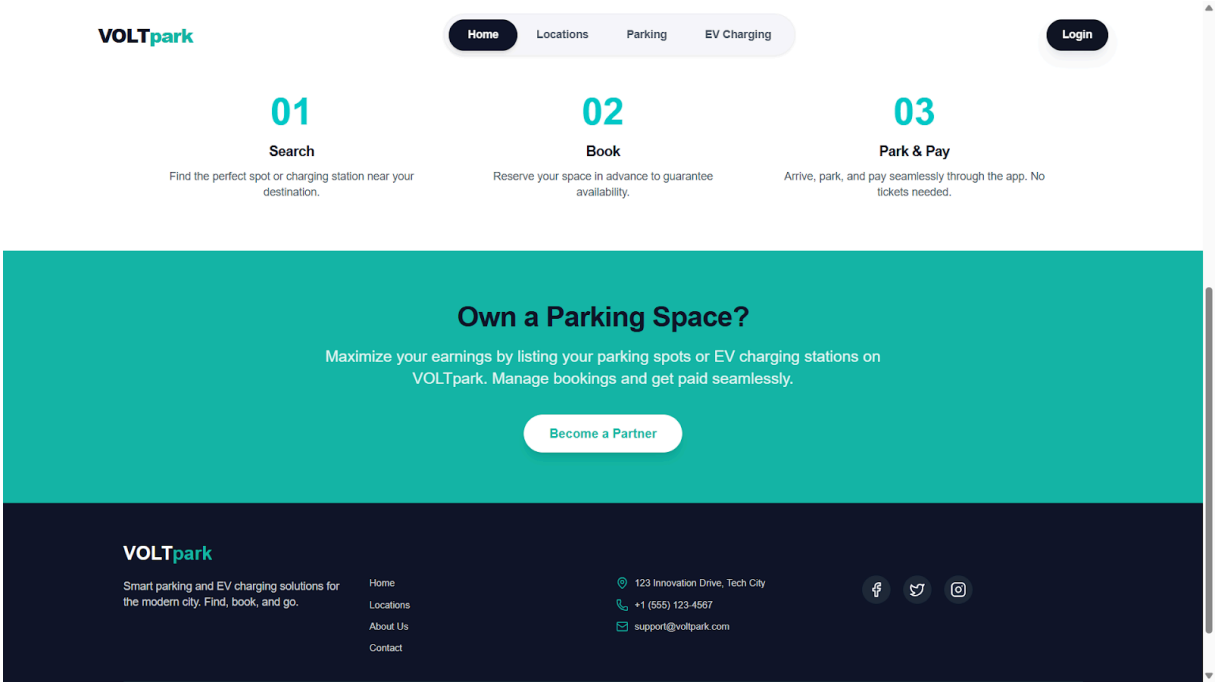

Fig 7.2 Landing page

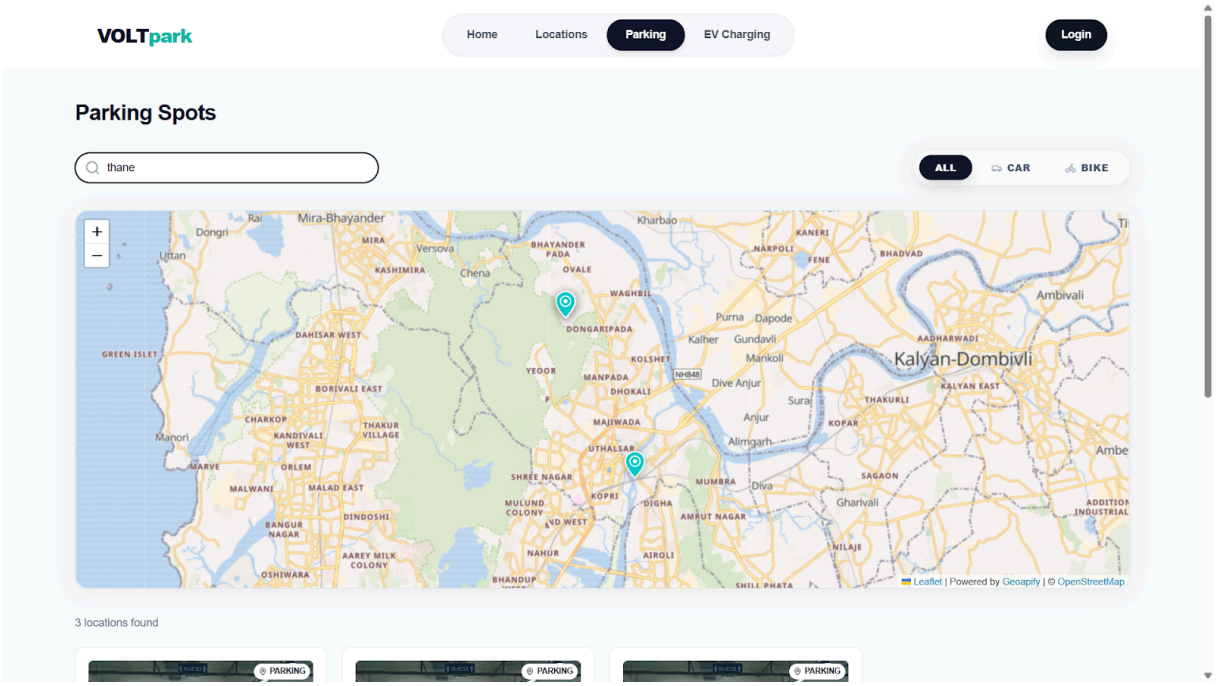

Fig 7.3 Google maps

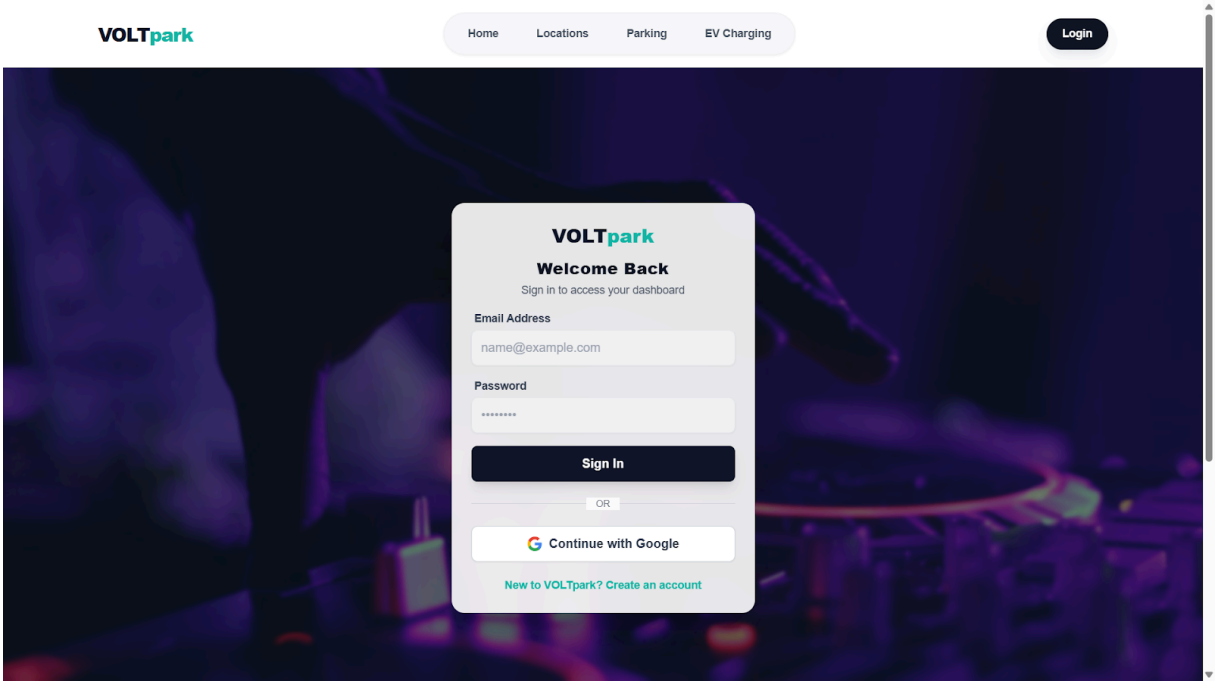

Fig 7.4 User login page

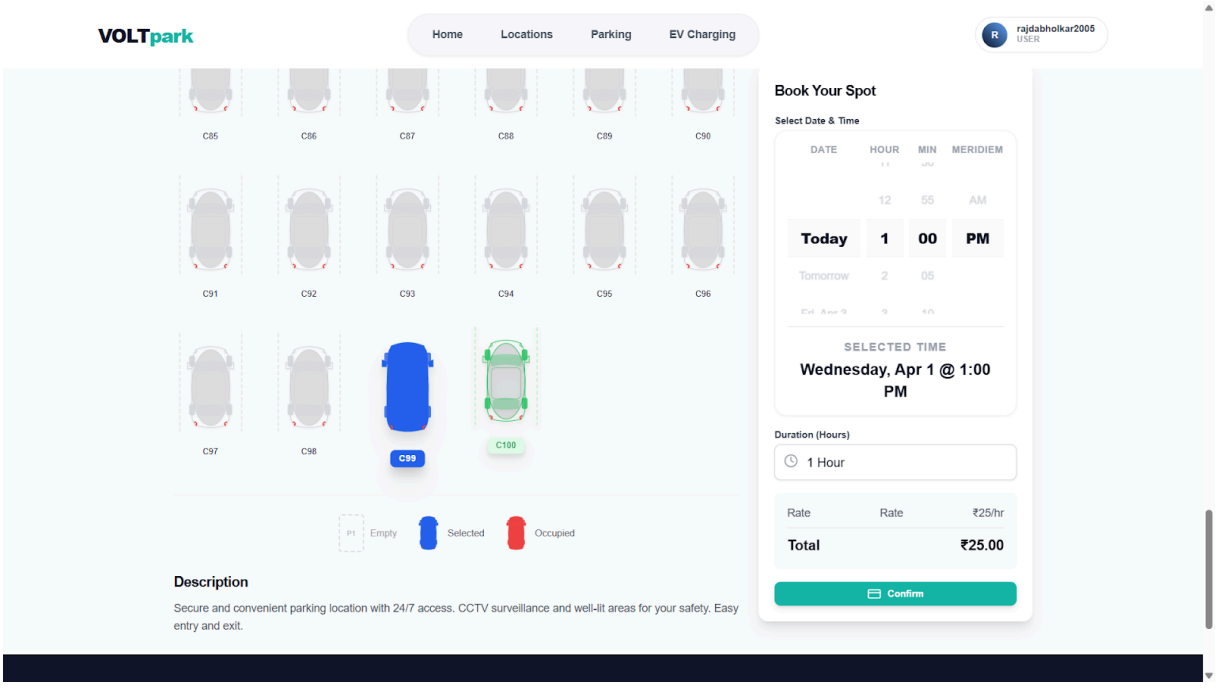

Fig 7.5 Booking slots

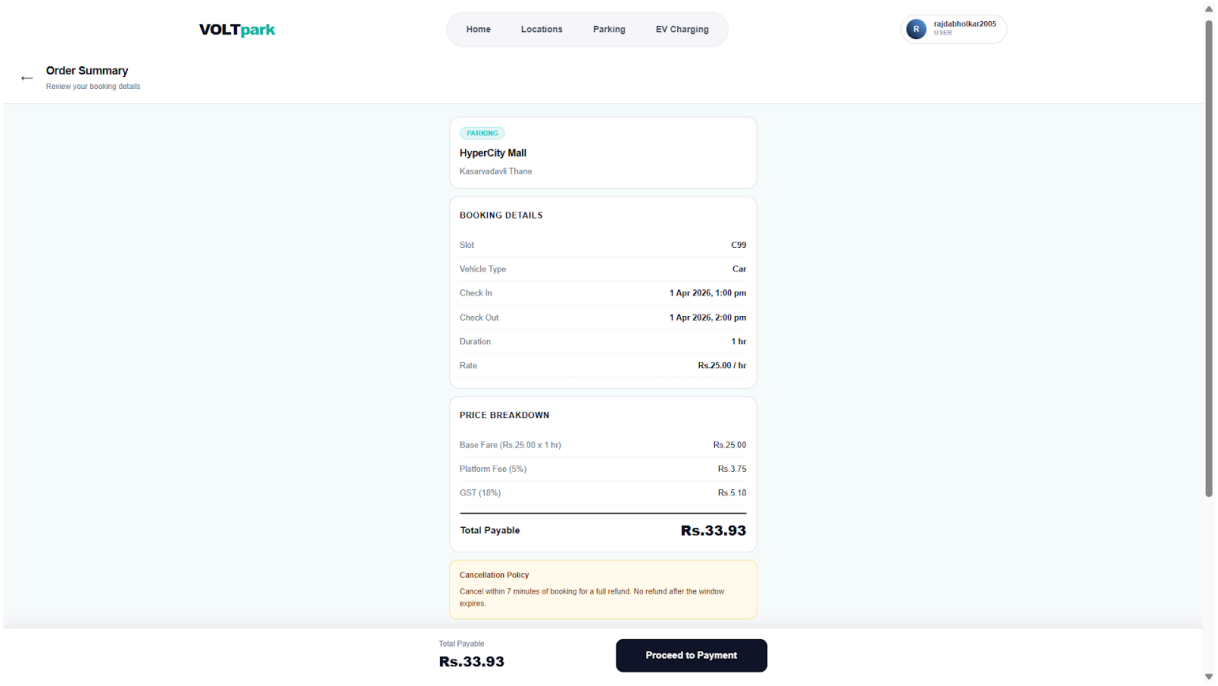

Fig 7.6 Payment breakdown

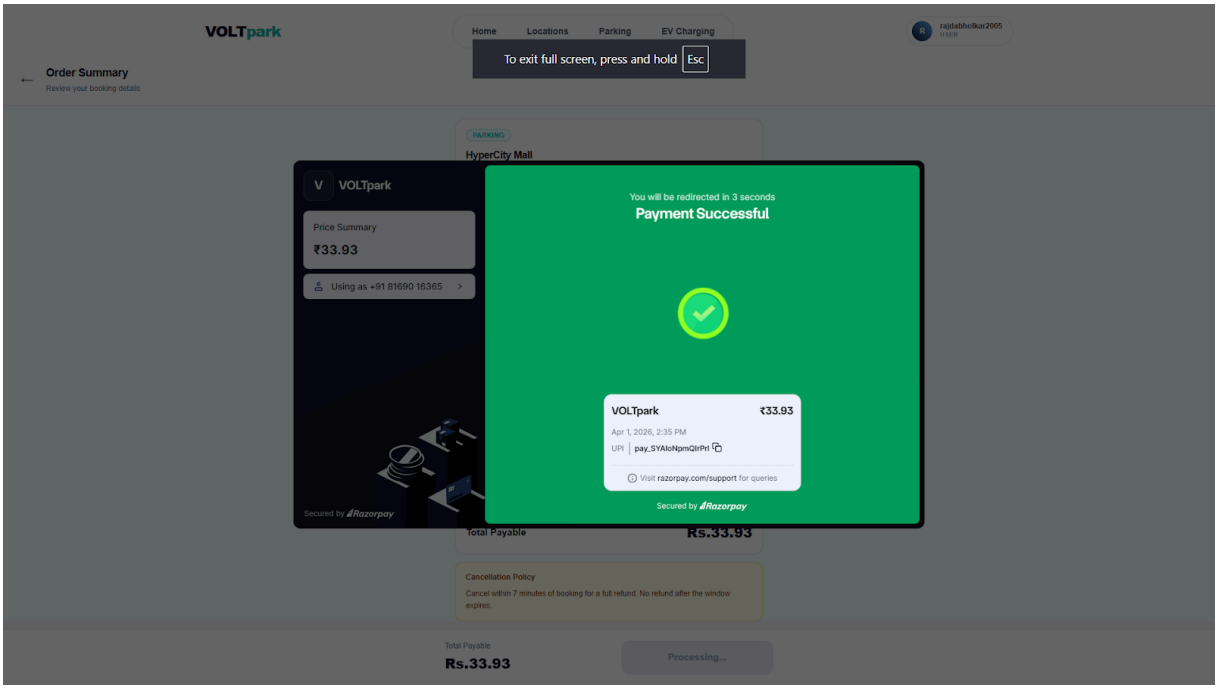

Fig 7.7 Payment Gateway

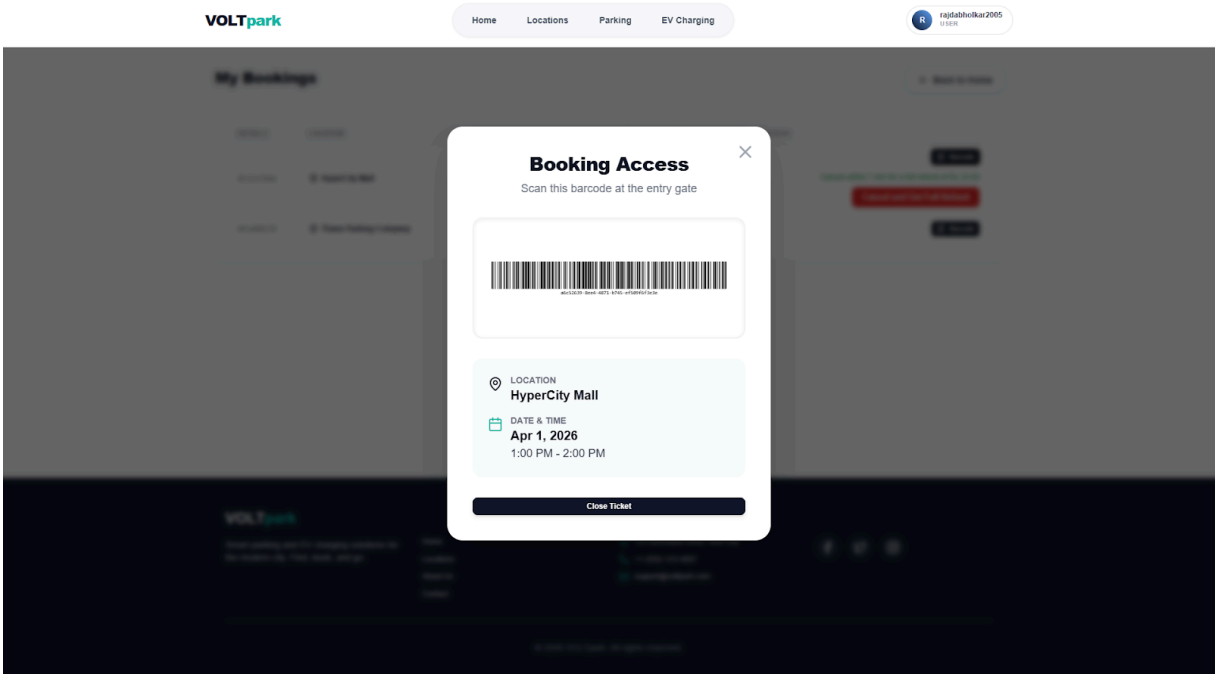

Fig 7.8 Barcode generation

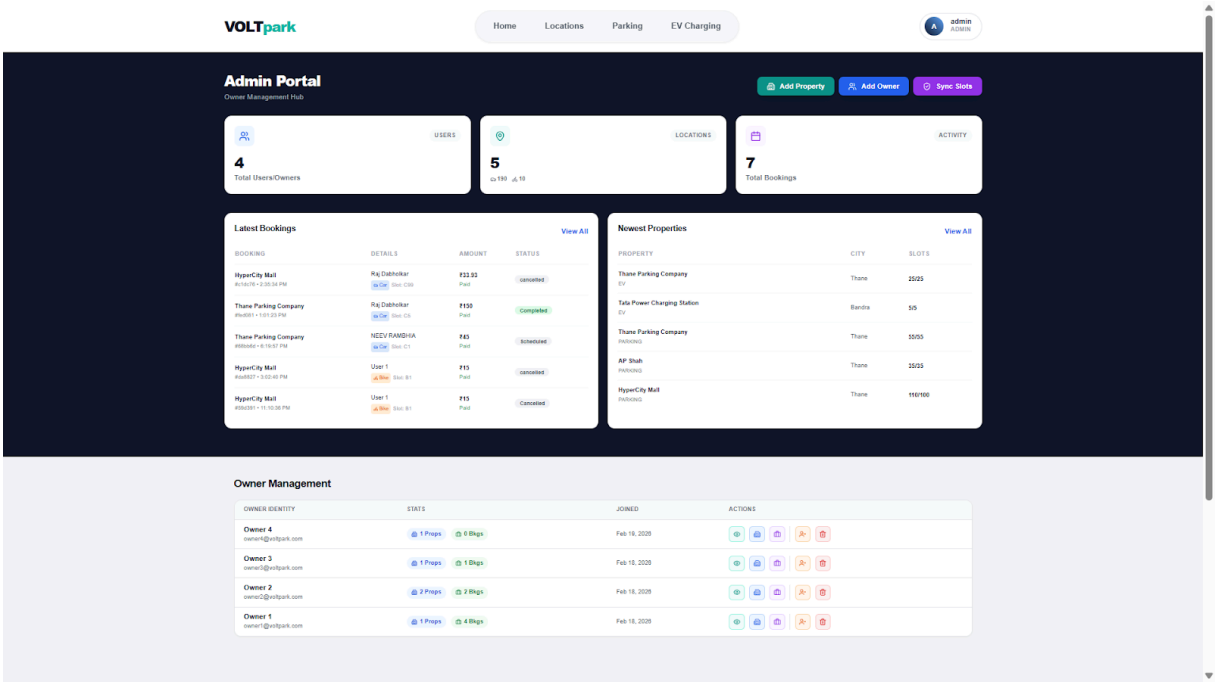

Fig 7.9 Owner dashboard

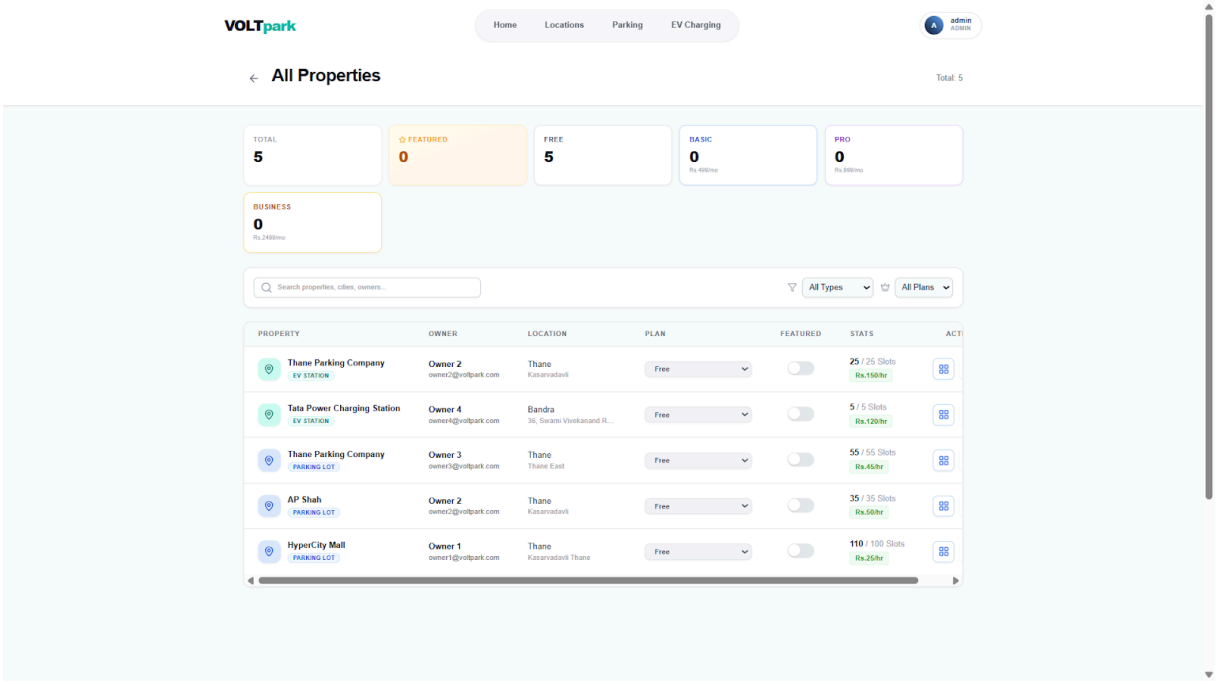

Fig 7.10 List of total properties

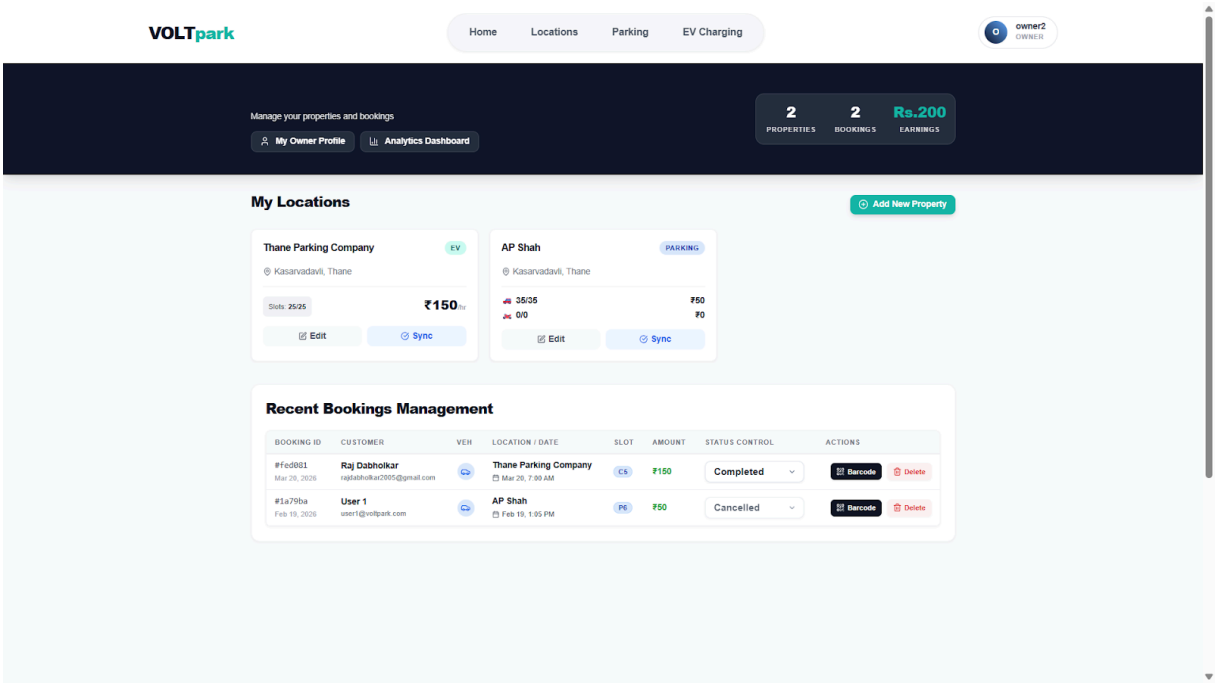

Fig 7.11 Admin Dashboard

Chapter 8

Conclusion and Future Work

8.Conclusion and Future Work

This chapter presents the overall conclusion of the VOLTpark system and outlines the potential future enhancements that can further improve its functionality and scalability.

8.1 Conclusion

The VOLTpark system successfully addresses the challenges associated with urban parking management and limited accessibility of electric vehicle charging infrastructure. The system provides a unified, web-based platform that enables users to search, view, and book parking and EV charging facilities in real time. By integrating multiple functionalities such as slot visualization, booking management, role-based dashboards, and digital payment processing, the platform offers a comprehensive solution to the problem of inefficient parking systems.

The implementation of real-time slot availability using backend services ensures accurate and up-to-date information for users, reducing the chances of double booking and improving overall user experience. The system also enables property owners to manage their parking infrastructure efficiently and monetize underutilized resources. The administrator module ensures proper control and monitoring of platform activities, maintaining system reliability and security.

The results obtained from system testing demonstrate significant improvements in operational efficiency, including reduced search time, increased slot utilization, and high booking success rates. The software-based approach eliminates the need for costly hardware infrastructure, making the system scalable and accessible for widespread adoption.

Overall, the VOLTpark system achieves its primary objective of providing an efficient, user-friendly, and scalable solution for smart parking and EV charging management, contributing to improved urban mobility and sustainable development.

8.2 Future Work

Although the VOLTpark system demonstrates strong performance and functionality, there are several areas where further improvements and enhancements can be implemented.

Future work includes the integration of IoT-based sensors to enable automated and highly accurate slot detection without manual updates. The implementation of dynamic pricing models based on demand, time, and location can further optimize resource utilization and revenue generation for property owners.

The system can be extended to include advanced features such as AI-based demand prediction, which can forecast parking availability and guide users more effectively. The development of a mobile application can improve accessibility and provide a better user experience for on-the-go users.

Further enhancements can include advanced analytics dashboards for administrators, enabling data-driven decision-making based on user behavior and system usage patterns. Integration with smart city infrastructure and expansion to multiple cities can increase the system's impact and scalability.

Additionally, improvements in system performance through cloud optimization, load balancing, and distributed architecture can ensure smooth operation under high user traffic conditions.

These enhancements will further strengthen the capabilities of the VOLTpark system and support its transition from a prototype to a fully deployable real-world solution.

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